FROM ARTIFICIAL INTELLIGENCE TO ORGANISM

Neuropeptides are known to regulate a variety of physiological processes, including: feeding and metabolism, sleep and circadian rhythm, stress, learning and memory, addiction and reward. In the model organism *Drosophila melanogaster*, these systems are well characterized and experimentally accessible. However, designing peptides with specific, desired biological functions remains challenging. At the same time, advances in artificial intelligence enable the design of novel peptides with tailored biological properties.

Your task for S arena this year is to design a peptide using artificial intelligence to modulate a selected biological process in a *Drosophila*.

Day 1

1. Select biological target (a neuropeptide system, receptor, or process).
2. Explain the relevance of the chosen target, known peptide ligands and their function.
3. Explain potential therapeutic relevance of your peptide (mimicry, antagonism..)
4. Is your peptide a "good peptide"?
    a. Synthesizability: *How easy is it to make? Focus on peptides with 6–10 amino acids; they are easier to model and synthesize.*
    b. Pharmacokinetics & Delivery: *How will it be administered to the fly/human (oral, intravenous, etc.)? Does it need to enter the cell to work? How does it cross the membrane (diffusion, net charge, hydrophobicity)? Does it require specific transporters to reach its destination?*
    c. Stability & Solubility: *How difficult is it to handle in a lab setting?*
    d. Affinity & Specificity: *How well does it bind to your target? (Is it reversible or irreversible? Does it accidentally recognize other receptors?)*
    e. Expected mechanism of action
5. AI Models for Peptide Design
    a. Dataset: *Curate your own dataset (5–10 known peptides).*
    b. Generative Process: *Describe how you would generate new sequences.*
    c. Evaluation: *Based on your examples, choose one peptide from your dataset. Describe how you would test its properties. What specific criteria (from Section 2) are you using to select the "winner"?*